MODERN PHYSICS – MIDTERM 2. RETAKE

Please write every problem on separate sheet of paper, and write your Neptun code and Name on every paper you submit. Make sure to write all the formulas that you evaluate, as well as your calculations, so you can get partial score even if there is an error. Give the answers with a precision of 3 significant digits, together with the units, unless otherwise specified. Note that if a problem is scored, e.g., X + Y + Z points, then *three* answers are expected. Read and start every part of the problems, you can still get partial score even if you don't have the earlier results.

Physical constants and useful formulas

$$m_e = 9.11 \times 10^{-31}\,\mathrm{kg} \qquad h = 6.626 \times 10^{-34}\,\mathrm{J\,s} = 4.136 \times 10^{-15}\,\mathrm{eV\,s} \qquad c = 3.00 \times 10^{8}\,\mathrm{m/s}$$

$$m_e c^2 = 511\,\mathrm{keV} \qquad hc = 1.986 \times 10^{-25}\,\mathrm{J\,m} = 1240\,\mathrm{eV\,nm} \qquad \frac{\hbar^2}{2m} = 3.810 \times 10^{-2}\,\mathrm{eV nm^2}$$

$$\gamma = \left(1 - \frac{v^2}{c^2}\right)^{-1/2} \qquad \lambda_C = \frac{h}{m_e c} = 2.43 \times 10^{-12}\,\mathrm{m} \qquad \lambda' - \lambda = \lambda_C(1 - \cos\theta)$$

1. Electronic configuration (1 + 2 + 3 = 6 points)

Consider the Titanium atom (Ti), whose *ground-state* electronic configuration is: [Ar] $4s^2$ $3d^2$.

(a) Draw the orbital energy diagram corresponding to the 4s and 3d subshells.

(b) Fill the orbitals with electrons according to *Hund's rules* and the *Pauli exclusion principle*.

(c) Using the resulting electronic configuration, determine the quantum numbers S, L, and J

Show your reasoning clearly.

Hint: according to Hund's rules, the minimum energy state is obtained by (1) maximizing S*, (2)* maximizing L*, and (3) for a less-than-half-filled shell (fulfilled for* Ti*)* minimizing J*.*

2. Compton scattering (3 + 3 + 4 = 10 points)

A photon with energy $E = 40\,\mathrm{keV}$ scatters from a free electron at rest.

(a) What is the wavelength λ of the incident photon?

(b) If the scattering angle is $\theta = 60°$, calculate the *red-shifted* wavelength λ' of the scattered photon.

(c) What is the maximum recoil kinetic energy (over all scattering angles) the electron can obtain in the process?

3. Stellar nucleosynthesis (1 + 2 + 2 + 1 = 6 points)

Consider the *hypothetical* nuclear reaction, in which Iron (Fe) fusion is used to synthesize Tellurium (Te):

$$^{56}_{26}\mathrm{Fe} + {}^{56}_{26}\mathrm{Fe} \rightarrow {}^{112}_{52}\mathrm{Te}$$

Using the tabulated atomic masses below

(a) Calculate the mass difference (in units of the atomic mass u)

(b) Estimate the $\mathcal{Q}$-value (in units of eV or MeV).

(c) Determine the minimum velocity $v_{\min}$ that the Fe atoms must have *in the center of mass frame* for the reaction to be energetically possible (ignoring the Coulomb barrier).

(d) Discuss briefly why such a reaction is *not* relevant for energy production in stellar environments.

Atomic masses: $M_{\mathrm{Fe}} = 55.934942$ u; $M_{\mathrm{Te}} = 111.917060$ u; $u = 931.5\,\mathrm{MeV/c^2}$.

4. Stern-Gerlach experiment for $S = 1$ (4 + 5 + 3 = 12 points)
Consider a Stern-Gerlach experiment with atoms having spin $S = 1$. The z basis is composed of the following orthonormal vectors:

$$|1\rangle = \begin{pmatrix}1\\0\\0\end{pmatrix} \qquad |0\rangle = \begin{pmatrix}0\\1\\0\end{pmatrix} \qquad |-1\rangle = \begin{pmatrix}0\\0\\1\end{pmatrix}$$

In this basis, the operators S_z and S_x are represented by

$$S_z = \hbar \begin{pmatrix}1&0&0\\0&0&0\\0&0&-1\end{pmatrix} \qquad \text{and} \qquad S_x = \frac{\hbar}{\sqrt{2}} \begin{pmatrix}0&1&0\\1&0&1\\0&1&0\end{pmatrix}.$$

For such operators

(a) Verify that the eigenvalues are $\{+\hbar, 0, -\hbar\}$ for both S_z and S_x.

(b) Find the *normalized* eigenstate $|+\rangle$ of S_x corresponding to eigenvalue $+\hbar$, and express it in the form

$$|+\rangle = \alpha\,|1\rangle + \beta\,|0\rangle + \gamma\,|-1\rangle\,,$$

where α, β, γ are numerical constants.

(c) We direct the beam of electrons with the $|+\rangle$ state into a Stern-Gerlach setup, with the field gradient pointing in the z direction. What are the intensities of the three beams observed relative to the intensity of the incoming beam?
Hint: Calculate the probabilities $P(+z)$, $P(0)$, and $P(-z)$ of the atoms in the $|+\rangle$ state pointing in the $+z$, 0, and $-z$ direction. Remember that $P(+z) + P(0) + P(-z) = 1$.

5. Variational Methods: Gaussian potential well (2 + 4 + 4 + 3 + 2 + 1 = 16 points)
Consider an electron in a one-dimensional attractive potential, described by the Hamiltonian

$$H = K + V = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} - V_0\, e^{-\beta x^2},$$

where $\hbar^2/(2m) = 3.810 \times 10^{-2}\,\mathrm{eVnm}^2$, $V_0 = 2\,\mathrm{eV}$ and $\beta = 2\,\mathrm{nm}^{-2}$.
Consider a variational wavefunction, which we choose to also have a Gaussian form, characterized by a single parameter α with nm^{-2} units, and a normalization constant N:

$$\psi(r) = N e^{-\alpha x^2/2}.$$

(a) Write the normalization condition and show that it is fulfilled for $N = \left(\frac{\alpha}{\pi}\right)^{\frac{1}{4}}$.
Hint: use the integral identity $\int_{-\infty}^{\infty} \mathrm{d}x\, e^{-ax^2} = \sqrt{\frac{\pi}{a}}$.

(b) Calculate the expression for the kinetic energy expectation value $\langle K\rangle_\psi$ that depends parametrically on α.
Hint: use the integral identity $\int_{-\infty}^{\infty} \mathrm{d}x\, x^2 e^{-ax^2} = \frac{\sqrt{\pi}}{2} a^{-\frac{3}{2}}$, and don't forget the normalization constant.

(c) Calculate the expression for the potential energy expectation value $\langle V\rangle_\psi$ that depends parametrically on α.

(d) Calculate the expression for the energy expectation value $E(\alpha) = \langle H\rangle_\psi = \langle K\rangle_\psi + \langle V\rangle_\psi$, and apply the variational principle to find an equation for α in the ground state.
Hint: bring the equation to the form $\alpha(\alpha+\beta)^3 = \gamma^2$ where $\gamma = 105.0\,\mathrm{nm}^{-4}$ is a combination of $\hbar$, m, V_0 and β.

(e) This is a complicated equation, but it can be solved in the limit where $\alpha \gg \beta$. Using the approximation $(\alpha+\beta)^3 \approx \alpha^3$, find the approximate solution α_0, calculate its numerical value, and verify that $\alpha_0 \gg \beta$.

(f) Calculate the approximate ground state energy $E_0 = E(\alpha_0)$.
Hint: as a reference value, the numerical solution for this potential yields $E_0 = -1.638\,\mathrm{eV}$.

OFFICIAL SOLUTIONS

1. Electronic configuration (1 + 2 + 3 = 6 points)
Consider the Titanium atom (Ti), whose *ground-state* electronic configuration is: [Ar] $4s^2$ $3d^2$.

(a) Draw the orbital energy diagram corresponding to this electronic configuration.

(b) Fill the orbitals with electrons according to *Hund's rules* and the *Pauli exclusion principle*.

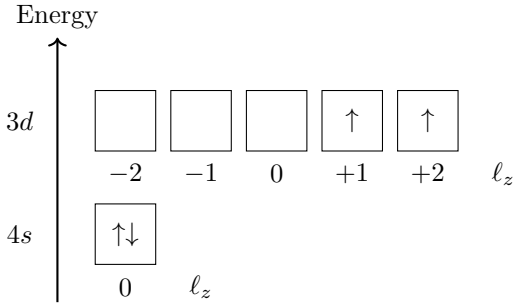
(c) Using the resulting electronic configuration, determine the quantum numbers S, L, and J

Show your reasoning clearly.

Hint: according to Hund's rules, the minimum energy state is obtained by (1) maximizing S*, (2)* maximizing L*, and (3) for a less-than-half-filled shell (fulfilled for* Ti*)* minimizing J*.*

Solution
The two key pieces of information are that Titanium has four valence electrons and that the 4s shell is lower in energy than the 3d shell.

(a) The corresponding energy diagram is shown below.



(b) The lower 4s shell is completely filled with two electrons with opposite spin (Pauli principle). The remaining two electrons occupy the two 3d orbitals corresponding with the highest values of $\ell_z = \{2, 1\}$, thus maximizing both the spin S (Hund I) and the orbital angular momentum L (Hund II).

(c) In this configuration, $S = 1$, $L = 3$. The total angular momentum can assume values $|L - S| \leq J \leq L + S$, and the ground state corresponds to $J = |L - S|$ (Hund III).

2. Compton scattering (3 + 3 + 4 = 10 points)
A photon with energy $E = 40\,\text{keV}$ scatters from a free electron at rest.

(a) What is the wavelength λ of the incident photon?

(b) If the scattering angle is $\theta = 60°$, calculate the *red-shifted* wavelength λ' of the scattered photon.

(c) What is the maximum recoil kinetic energy (over all scattering angles) the electron can obtain in the process?

Solution

(a) The wavelength associated with the photon is given by

$$\lambda = \frac{h}{p} = \frac{hc}{pc} = \frac{hc}{E} = 0.0310\,\text{nm}$$

(b) The change in the wavelength due to Compton scattering is given by

$$\lambda' - \lambda = \frac{h}{m_e c}\left(1 - \cos(\theta)\right)$$

so that for $\theta = 60°$ one finds

$$\lambda' - \lambda = \frac{h}{m_e c}\frac{1}{2} = \frac{hc}{m_e c^2}\frac{1}{2} = 0.001\,21\,\text{nm}$$

Since $\lambda' > \lambda$ (always) $f' < f$, and the photon is indeed *red-shifted* with $\lambda' = 0.0322\,\text{nm}$

(c) The total energy of the systems $e^- + \gamma$ before the scattering is given by

$$E = hf + m_e c^2$$

and after the scattering is given by

$$E' = hf' + m_e c^2 + K$$

Hence, by the conservation of energy $E = E'$, and the recoil kinetic energy of the electron is therefore equal to the energy lost by the photon

$$K = h(f - f')$$

Hence, the recoil kinetic energy is maximized when the photon is *back-scattered*, i.e., for $\theta = \pi$, for which

$$\lambda' - \lambda = \frac{hc}{m_e c^2}\,2$$

corresponding to $\lambda' = 0.035\,85\,\text{nm}$

$$\Delta E_\gamma = h(f - f') = hc\left(\frac{1}{\lambda} - \frac{1}{\lambda'}\right)$$

resulting in a kinetic energy

$$K = 5.414\,\text{keV}.$$

3. Stellar nucleosynthesis (1 + 2 + 2 + 1 = 6 points)
Consider the *hypothetical* nuclear reaction, in which Iron (Fe) fusion is used to synthesize Tellurium (Te):

$$^{56}_{26}\mathrm{Fe} + ^{56}_{26}\mathrm{Fe} \rightarrow ^{112}_{52}\mathrm{Te}$$

Using the tabulated atomic masses below

(a) Calculate the mass difference (in units of the atomic mass u)

(b) Estimate the $\mathcal{Q}$-value (in units of eV or MeV).

(c) What is the minimum velocity $v_{\min}$ that the Fe atoms must have *in the center of mass frame* for the reaction to be energetically possible? (Ignore the Coulomb barrier.)

(d) Discuss briefly why such a reaction is <u>not</u> relevant for energy production in stellar environments.

Atomic masses: $M_{\mathrm{Fe}} = 55.934942$ u; $M_{\mathrm{Te}} = 111.917060$ u; $u = 931.5\,\mathrm{MeV/c^2}$.

Solution

(a) The initial mass is given by

$$2M_{\mathrm{Fe}} = 2 \times 55.934942 = 111.869884\,\mathrm{u}.$$

whereas the final mass is just

$$M_{\mathrm{Te}} = 111.917060\,\mathrm{u}.$$

thus yielding the following mass difference

$$\Delta M = -0.047176\,\mathrm{u}.$$

(b) The $\mathcal{Q}$-value is given by

$$\mathcal{Q} = \Delta M c^2 = -43.9\,\mathrm{MeV} < 0.$$

(c) For the reaction to be energetically possible (neglecting the Coulomb barrier), it must be that

$$K \geq |\mathcal{Q}|, \qquad \text{or} \qquad K_{\min} = |\mathcal{Q}|.$$

In the center of mass reference frame, conservation of momentum requires that $\vec{p}_1 + \vec{p}_2 = 0$, so that the Fe atoms have the same momentum (and velocity, and kinetic energy). Therefore

$$E = K_{\min} + 2M_{\mathrm{Fe}}c^2 = \gamma\left(2M_{\mathrm{Fe}}c^2\right)$$

from which one derived $\gamma \simeq 1.00042$, where enough significant digits are included to appreciate that $\gamma > 1$, and

$$v = \beta c = \sqrt{1 - \frac{1}{\gamma^2}}\, c \simeq 0.029\, c \ll c.$$

Note that since $|\mathcal{Q}| = K_{\min} \ll E_0$, a *non-relativistic* approximation for the kinetic is also reasonable, i.e.,

$$2\left(\frac{1}{2}M_{\mathrm{Fe}}v^2\right) = K_{\min} = |\mathcal{Q}|$$

yielding

$$v = \sqrt{\frac{|\mathcal{Q}|}{M_{\mathrm{Fe}}c^2}}\, c \simeq 0.029\, c.$$

(d) Such reactions are not relevant for stellar energy production because nuclei heavier than Fe have lower binding energy per nucleon, so their fusion is *endothermic* ($\mathcal{Q} < 0$). As a result, they require an external energy input and cannot release energy to sustain stellar processes. Additionally, the Coulomb repulsion between very heavy nuclei strongly suppresses their fusion in stellar environments.

4. Stern-Gerlach exercise for $S = 1$ (4 + 5 + 3 = 12 points)
Consider a Stern-Gerlach experiment with atoms having spin $S = 1$. The z basis is composed of the following orthonormal vectors:

$$|1\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \qquad |0\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \qquad |-1\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

In this basis, the operators S_z and S_x are represented by

$$S_z = \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \qquad \text{and} \qquad S_x = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

For such operators

(a) Verify that the eigenvalues are $\{+\hbar, 0, -\hbar\}$ for both S_z and S_x.

(b) Find the *normalized* eigenstate $|+\rangle$ of S_x corresponding to eigenvalue $+\hbar$, and express it in the form

$$|+\rangle = \alpha\,|1\rangle + \beta\,|0\rangle + \gamma\,|-1\rangle\,.$$

(c) We direct the beam of electrons with the $|+\rangle$ state into a Stern-Gerlach setup, with the field gradient pointing in the z direction. What are the intensities of the three beams observed relative to the intensity of the incoming beam?
Hint: Calculate the probabilities $P(+z)$, $P(0)$, and $P(-z)$ of the atoms in the $|+\rangle$ state pointing in the $+z$, 0, and $-z$ direction. Remember that $P(+z) + P(0) + P(-z) = 1$.

Solution

(a) The S_z matrix is diagonal and, by definition, the diagonal elements correspond to the eigenvalues.
For S_x we solve the secular equation given by

$$0 = \det\left(S_x - \lambda I\right) = \begin{vmatrix} -\lambda & \hbar/\sqrt{2} & 0 \\ \hbar/\sqrt{2} & -\lambda & \hbar/\sqrt{2} \\ 0 & \hbar/\sqrt{2} & -\lambda \end{vmatrix} = \left(-\lambda^3 + \hbar^2\lambda\right) = \lambda(\hbar^2 - \lambda^2) = 0$$

which has solutions $\lambda_0 = 0$, and $\lambda_\pm = \pm\hbar$.

(b) To find the $|+\rangle$ eigenstate, we need to solve

$$S_x\,|+\rangle = \hbar\,|+\rangle$$

which after simplifying $\hbar$ yields

$$\begin{pmatrix} -1 & 1/\sqrt{2} & 0 \\ 1/\sqrt{2} & -1 & 1/\sqrt{2} \\ 0 & 1/\sqrt{2} & -1 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = 0.$$

with a normalized solution given by

$$|+\rangle = \frac{1}{2} \begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix} = \frac{1}{2}\,|1\rangle + \frac{\sqrt{2}}{2}\,|0\rangle + \frac{1}{2}\,|-1\rangle\,.$$

(c) After passing through the Stern-Gerlach setup, the probabilities correspond to

$$P(+z) = |\langle 1|+\rangle|^2 = |\alpha|^2 = \frac{1}{4}$$
$$P(0) = |\langle 0|+\rangle|^2 = |\beta|^2 = \frac{1}{2}$$
$$P(-z) = |\langle -1|+\rangle|^2 = |\gamma|^2 = \frac{1}{4}$$

which satisfy $P(+z) + P(0) + P(-z) = 1$.

5. Variational Methods: Gaussian potential well (2 + 4 + 4 + 3 + 2 + 1 = 16 points)
Consider an electron in a one-dimensional attractive potential, described by the Hamiltonian

$$H = K + V = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} - V_0\, e^{-\beta x^2},$$

where $\hbar^2/(2m) = 3.810 \times 10^{-2}\,\mathrm{eVnm}^2$, $V_0 = 2\,\mathrm{eV}$ and $\beta = 2\,\mathrm{nm}^{-2}$.
Consider a variational wavefunction, which we choose to also have a Gaussian form, characterized by a single parameter α with nm^{-2} units, and a normalization constant N:

$$\psi(r) = Ne^{-\alpha x^2/2}.$$

(a) Write the normalization condition and show that it is fulfilled for $N = \left(\frac{\alpha}{\pi}\right)^{\frac{1}{4}}$.
Hint: use the integral identity $\int_{-\infty}^{\infty} \mathrm{d}x\, e^{-ax^2} = \sqrt{\frac{\pi}{a}}$.

(b) Calculate the expression for the kinetic energy expectation value $\langle K\rangle_\psi$ that depends parametrically on α.
Hint: use the integral identity $\int_{-\infty}^{\infty} \mathrm{d}x\, x^2 e^{-ax^2} = \frac{\sqrt{\pi}}{2}a^{-\frac{3}{2}}$, *and don't forget the normalization constant.*

(c) Calculate the expression for the potential energy expectation value $\langle V\rangle_\psi$ that depends parametrically on α.

(d) Calculate the expression for the energy expectation value $E(\alpha) = \langle H\rangle_\psi = \langle K\rangle_\psi + \langle V\rangle_\psi$, and apply the variational principle to find an equation for α in the ground state.
Hint: bring the equation to the form $\alpha(\alpha+\beta)^3 = \gamma^2$ *where* $\gamma = 105.0\,\mathrm{nm}^{-4}$ *is a combination of* $\hbar$, m, V_0 *and* β.

(e) This is a complicated equation, but it can be solved in the limit where $\alpha \gg \beta$. Using the approximation $(\alpha+\beta)^3 \approx \alpha^3$, find the approximate solution α_0, calculate its numerical value, and verify that $\alpha_0 \gg \beta$.

(f) Calculate the approximate ground state energy $E_0 = E(\alpha_0)$.
Hint: as a reference value, the numerical solution for this potential yields $E_0 = -1.638\,\mathrm{eV}$.

Solution

(a) The normalization condition reads

$$1 = \int_{-\infty}^{\infty} \mathrm{d}x\, \psi^*(x)\psi(x) = N^2 \int_{-\infty}^{\infty} \mathrm{d}x\, e^{-\alpha x^2} = N^2 \left(\frac{\pi}{\alpha}\right)^{1/2}$$

thus yielding

$$N = \left(\frac{\alpha}{\pi}\right)^{\frac{1}{4}}$$

(b) The expectation value of the kinetic energy is given by

$$\langle\psi|\, K\,|\psi\rangle = -\frac{\hbar^2}{2m}\left(\frac{\alpha}{\pi}\right)^{1/2}\int_{-\infty}^{\infty}\mathrm{d}x\, e^{-\alpha x^2/2}\left(\frac{\partial^2}{\partial x^2}e^{-\alpha x^2/2}\right) = -\frac{\hbar^2}{2m}\left(\frac{\alpha}{\pi}\right)^{1/2}\int_{-\infty}^{\infty}\mathrm{d}x\, e^{-\alpha x^2}\left[-\alpha\left(1-\alpha x^2\right)\right] = \frac{\hbar^2}{4m}\alpha$$

(c) The expectation value of the potential energy is given by

$$\langle\psi|\, V\,|\psi\rangle = -V_0\left(\frac{\alpha}{\pi}\right)^{1/2}\int_{-\infty}^{\infty}\mathrm{d}x\, e^{-(\alpha+\beta)x^2} = -V_0\frac{\alpha^{1/2}}{(\alpha+\beta)^{1/2}}$$

(d) The parametric expression for ground state energy reads

$$E(\alpha) = \frac{\hbar^2}{4m}\alpha - V_0\sqrt{\frac{\alpha}{(\alpha+\beta)}},$$

for which one can verify that each term has the proper physical dimensions.
The variational principle implies that

$$\frac{d}{d\alpha}E(\alpha) = \frac{\hbar^2}{4m} - V_0\frac{1}{2}\left[\frac{1}{\alpha^{1/2}(\alpha+\beta)^{1/2}} - \frac{\alpha^{1/2}}{(\alpha+\beta)^{3/2}}\right] = 0$$

By rearranging the terms and squaring both the left- and right-hand sides one gets

$$\left(\frac{\hbar^2}{2mV_0}\right)^2 = \left[\frac{1}{\alpha(\alpha+\beta)} + \frac{\alpha}{(\alpha+\beta)^3} - \frac{2}{(\alpha+\beta)^2}\right] = \frac{\beta}{\alpha(\alpha+\beta)^3}.$$

Therefore, the equation reduces to

$$\alpha(\alpha+\beta)^3 = \gamma^2,$$

where

$$\gamma^2 = \beta^2 \left(\frac{2mV_0}{\hbar^2}\right)^2.$$

(e) In the limit $\alpha \gg \beta$ the expression above reduces to

$$\alpha^4 = \gamma^2$$

which admits the solution

$$\alpha_0 = \sqrt{\gamma} = 10.25\,\mathrm{nm}^{-2} \gg \beta.$$

(f) Substituting the optimal value $\alpha = \alpha_0$ in the parametric expression for the ground state energy, one finds

$$E_0 = E(\alpha_0) = -1.634\,\mathrm{eV}$$

which differs from the numerical value by less than 1 %.